High Voltage Low Power Consumption LDO MD7218 Series

CMOS Voltage Regulator 1A

MD7218 series is a high voltage (up to 18V) ultra-low quiescent current low dropout voltage regulator (LDO) manufactured in CMOS processes. It can deliver up to 1A of current while consuming only 1.6uA of quiescent current. It consists of a reference voltage generator, an error amplifier, a current foldback circuit, and a phase compensation circuit plus a driver transistor. The MD7218 series is designed specifically for applications where very-low I_Q is a critical parameter. This device maintains low quiescent current consumption even in dropout mode to further increase the battery life.

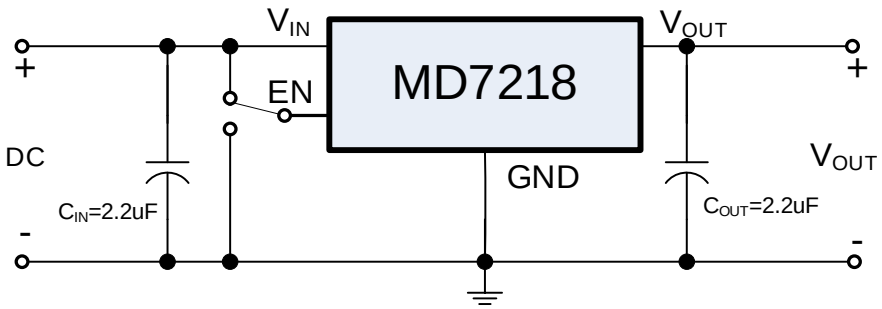
■ Features:

- Ultra-low Quiescent Current: 1.6uA
- Maximum Input Voltage: 18V
- Output Voltage Highly Accurate: ±2%
- Maximum Output Current: 1A
- Dropout Voltage: 1V@I_{OUT}=1A
- Temperature Stability: ±50ppm/℃
- ON/OFF Logic = Enable High
- Protections Circuits: Current Limiter, Foldback, Thermal shutdown
- Output Capacitor: Low ESR Ceramic Capacitor Compatible

■ Applications:

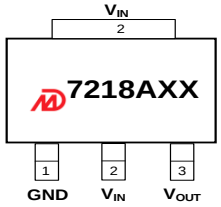
- Smart wearer
- Long-life battery-powered devices
- Portable mobile devices, such as mobile phones, cameras, and so on
- Wireless communication equipment

■ Typical Application:

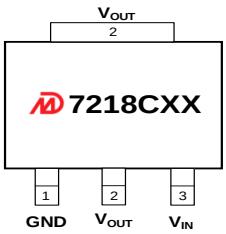


■ Pin Configuration (Top View):

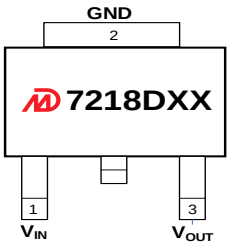
SOT89-3L(A_Type)
Top View



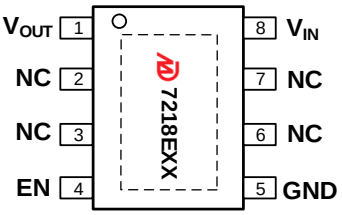
SOT223-3L(C_Type)
Top View



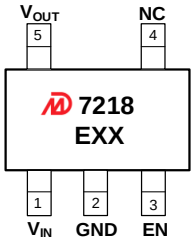
TO252-2L
Top View



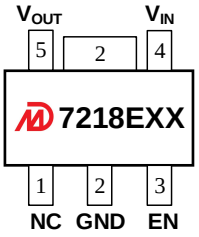
ESOP8
Top View



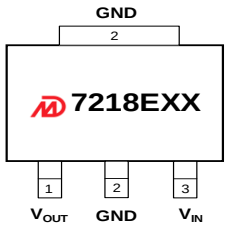
SOT23-5L
Top View



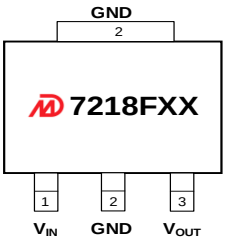
SOT89-5L
Top View



SOT89-3L(E_Type)
Top View



SOT223-3L(F_Type)
Top View



■ Product Selections

Product Name	VOUT (V)	Package	Ordering Name	Marking	Package Information
MD7218A30	3.0	SOT89-3L	MD7218A30PA1	7218A30	Tape and Reel, 1000pcs
MD7218A33	3.3	SOT89-3L	MD7218A33PA1	7218A33	
MD7218A36	3.6	SOT89-3L	MD7218A36PA1	7218A36	
MD7218A38	3.8	SOT89-3L	MD7218A38PA1	7218A38	
MD7218A40	4.0	SOT89-3L	MD7218A40PA1	7218A40	
MD7218A50	5.0	SOT89-3L	MD7218A50PA1	7218A50	
MD7218AC0	12.0	SOT89-3L	MD7218AC0PA1	7218AC0	
MD7218EC0	12.0	SOT89-3L	MD7218EC0PA1	7218EC0	
MD7218E30	3.0	SOT89-5L	MD7218E30PC1	7218E30	
MD7218E33	3.3	SOT89-5L	MD7218E33PC1	7218E33	
MD7218E36	3.6	SOT89-5L	MD7218E36PC1	7218E36	
MD7218E38	3.8	SOT89-5L	MD7218E38PC1	7218E38	
MD7218E40	4.0	SOT89-5L	MD7218E40PC1	7218E40	
MD7218E50	5.0	SOT89-5L	MD7218E50PC1	7218E50	

MD7218C18	1.8	SOT223-3L	MD7218C18YA2	7218C18	Tape and Reel, 2500pcs
MD7218C30	3.0	SOT223-3L	MD7218C30YA2	7218C30	
MD7218C33	3.3	SOT223-3L	MD7218C33YA2	7218C33	
MD7218C36	3.6	SOT223-3L	MD7218C36YA2	7218C36	
MD7218C40	4.0	SOT223-3L	MD7218C40YA2	7218C40	
MD7218C50	5.0	SOT223-3L	MD7218C50YA2	7218C50	
MD7218CC0	12.0	SOT223-3L	MD7218CC0YA2	7218CC0	
MD7218F33	3.3	SOT223-3L	MD7218F33YA2	7218F33	
MD7218F50	5.0	SOT223-3L	MD7218F50YA2	7218F50	
MD7218D30	3.0	TO252-2L	MD7218D30UA2	7218D30	Tape and Reel, 2500pcs
MD7218D33	3.3	TO252-2L	MD7218D33UA2	7218D33	
MD7218D36	3.6	TO252-2L	MD7218D36UA2	7218D36	
MD7218D40	4.0	TO252-2L	MD7218D40UA2	7218D40	
MD7218D50	5.0	TO252-2L	MD7218D50UA2	7218D50	
MD7218D60	6.0	TO252-2L	MD7218D60UA2	7218D60	
MD7218D80	8.0	TO252-2L	MD7218D80UA2	7218D80	
MD7218DC0	12.0	TO252-2L	MD7218DC0UA2	7218DC0	
MD7218E30	3.0	ESOP8	MD7218E30SF4	7218E30	Tape and Reel, 4000pcs
MD7218E33	3.3	ESOP8	MD7218E33SF4	7218E33	
MD7218E36	3.6	ESOP8	MD7218E36SF4	7218E36	
MD7218E40	4.0	ESOP8	MD7218E40SF4	7218E40	
MD7218E50	5.0	ESOP8	MD7218E50SF4	7218E50	
MD7218EC0	12.0	ESOP8	MD7218EC0SF4	7218EC0	
MD7218E30	3.0	SOT23-5L	MD7218E30QC3	7218E30	Tape and Reel, 3000pcs
MD7218E33	3.3	SOT23-5L	MD7218E33QC3	7218E33	
MD7218E36	3.6	SOT23-5L	MD7218E36QC3	7218E36	
MD7218E40	4.0	SOT23-5L	MD7218E40QC3	7218E40	
MD7218E50	5.0	SOT23-5L	MD7218E50QC3	7218E50	

Notes:

1* Customer can request to customize the output voltage ranged from 1.8V to 12V, if desired voltage is not found in the selections.

2* Customer can request customization of package choice.

3* Please pay attention to the MARKING of the product package type.

■ Absolute Maximum Ratings (Unless otherwise indicated: T_a=25℃)

PARAMETER	SYMBOL	RATINGS		UNITS
Input Voltage	V_{IN}	-0.3 ~ 20		V
Output Voltage	V_{OUT}	Vss-0.3 ~ VIN+0.3V		
Power Dissipation	P_D	SOT89-3L	1000	mW
		SOT89-5L	1000	
		SOT223-3L	1500	
		TO252-2L	1800	
		ESOP8	1800	
		SOT23-5L	250	
Thermal Resistance	$R_{\theta JA}$	SOT89-3L	100	℃/W
		SOT89-5L	100	
		SOT223-3L	66	
		TO252-2L	55	
		ESOP8	80	
		SOT23-5L	180	
Operating Ambient Temperature	T_{opr}	-40 ~ +85		℃
Storage Temperature	T_{stg}	-40 ~ +125		
ESD Protection	ESD HBM	4000		V

Note: Stresses above those listed under “Absolute Maximum Ratings” may cause permanent damage to the device.

■ Electrical Characteristics:

MD7218 Series (Unless otherwise indicated：T_a=25℃)

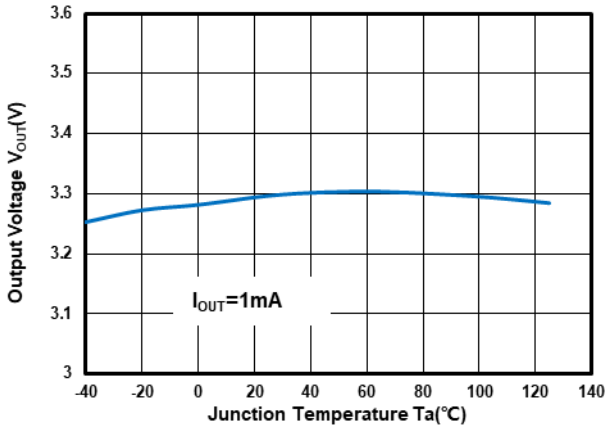
PARAMETER	SYMBOL	CONDITIONS		MIN.	TYP.	MAX.	UNIT
Output Voltage*1	$V_{OUT(S)}$	V_{IN}= $V_{OUT(S)}$+2V，I_{OUT}=1mA		$V_{OUT(S)}$×0.98	$V_{OUT(S)}$	$V_{OUT(S)}$×1.02	V
Dropout Voltage*2	V_{DROP}	$V_{OUT(S)}$=3.3V	I_{OUT}=1mA		3	8	mV
			I_{OUT}=1A		1000	1300	
Line Regulation	$\frac{\Delta V_{OUT}}{\Delta V_{IN} \bullet V_{OUT(s)}}$	$V_{OUT(S)}$+2V≤V_{IN}≤18V I_{OUT} =1mA			0.01	0.02	%/V
Load Regulation	ΔV_{OUT2}	V_{IN}=$V_{OUT(S)}$+2V 1mA≤I_{OUT}≤1A	$V_{OUT(S)}$≤5.0V		80		mV
			$V_{OUT(S)}$>5.0V		90		
Temperature Stability	$\frac{\Delta V_{OUT}}{\Delta T_a \bullet V_{OUT(s)}}$	V_{IN} = $V_{OUT(S)}$+2V，I_{OUT}=1mA -40℃≤T_a≤125℃			±50		ppm/℃
GND Current	I_{GND}	no load	$V_{OUT(S)}$≤5.0V	1	1.6	2	μA
			$V_{OUT(S)}$>5.0V	1.5	2.3	3	
		I_{OUT}=100mA			420		
Shutdown Current	I_{SHUT}	V_{IN}=18V, V_{EN}=0			0.1	1	
Input Voltage	V_{IN}	---		2.2		18	V
Maximum Output Current	I_{OUTMAX}			1			A
Current Limit*3	I_{LIM}	V_{IN}= $V_{OUT(S)}$+2V, V_{OUT} = 0.95 ×$V_{OUT(S)}$			1.5		
Power Supply Rejection Ratio*4	PSRR	f=10Hz, I_{OUT}=10mA			75		dB
		f=100Hz, I_{OUT}=10mA			80		
		f=1kHz, I_{OUT}=10mA			65		
Short Circuit Current*5	I_{SHORT}	V_{IN}=$V_{OUT(S)}$+2.0V V_{OUT}=0V			30		mA
EN 'H' Level Voltage	V_{ENH}			1.5		18	V
EN 'L' Level Voltage	V_{ENL}			0		0.5	
EN 'H' Level Current	I_{ENH}	V_{IN}=18V, V_{EN}=V_{IN}		-0.1		0.1	uA
EN 'L' Level Voltage	I_{ENL}	V_{IN}=18V, V_{EN}=0		-0.1		0.1	
Over Temperature Protection	OTP	I_{OUT}=1mA			150		℃

Notes:

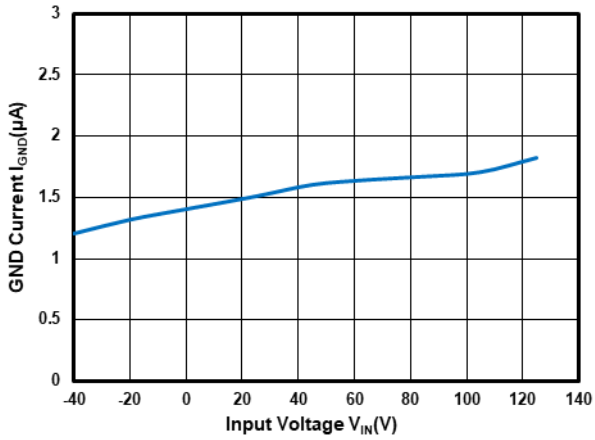
1. $V_{OUT(S)}$: Output voltage when V_{IN}=V_{OUT}+2V, I_{OUT}=1 mA.
2. V_{DROP}=V_{IN1} -（$V_{OUT(S)}$× 0.98）where V_{IN1} is the input voltage when V_{OUT} = $V_{OUT(S)}$× 0.98.
3. I_{LIM}: Output current when V_{IN}=$V_{OUT(S)}$+2V and V_{OUT} = 0.95*$V_{OUT(S)}$.
4. PSRR was measured for $V_{OUT(S)}$ = 3.3V and V_{IN}= 5.3V.
5. VOUT pin should be shorted to GND pin, and the impedance between them is less than 0.1 ohm

■ Typical Performance Characteristics:

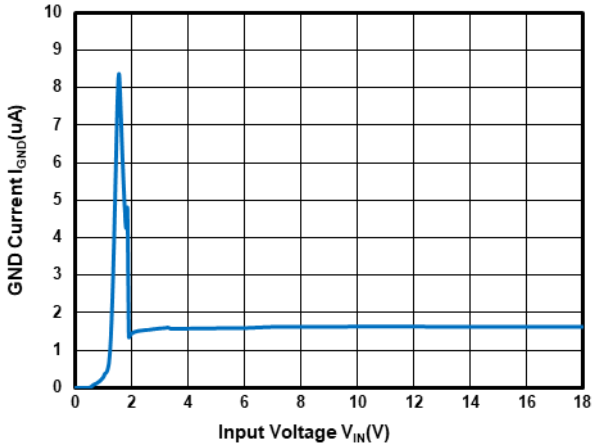
Test Conditions: V_{IN}=V_{OUT}+2.0V, C_{IN}=2.2μF, C_{OUT}=2.2μF, unless otherwise indicated.



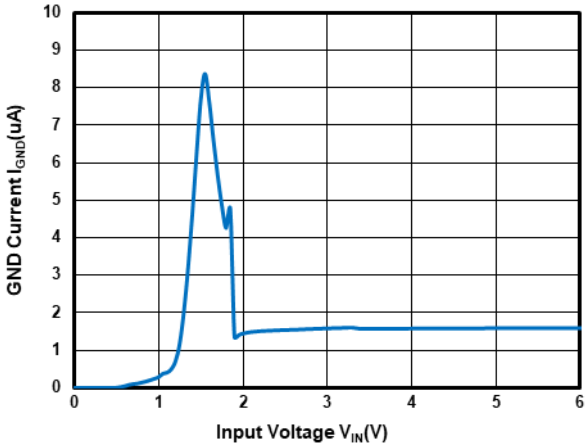
Output Voltage vs Temperature at V_{OUT}=3.3V



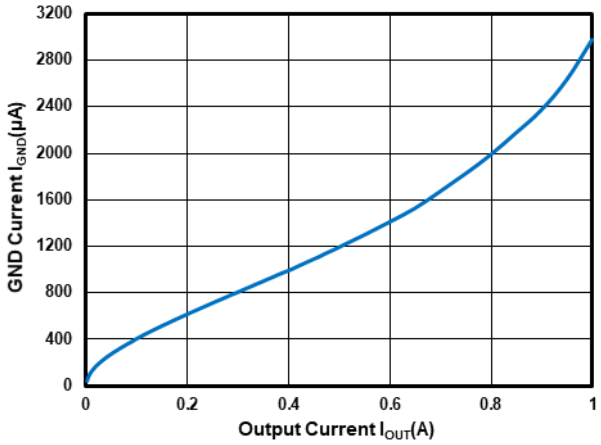
GND Current vs Temperature at V_{OUT}=3.3V



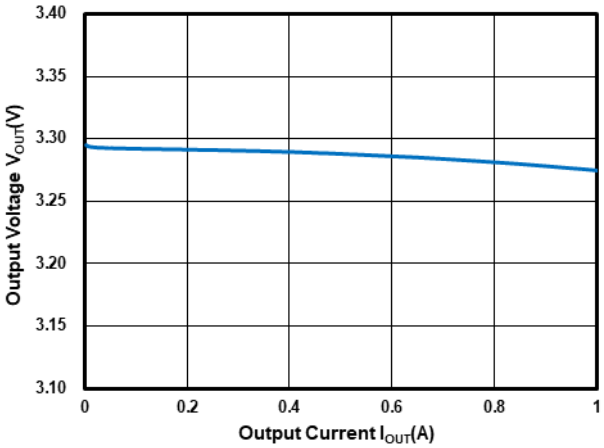
GND Current vs Input Voltage at V_{OUT}=3.3V



GND Current vs Input Voltage at V_{OUT}=3.3V



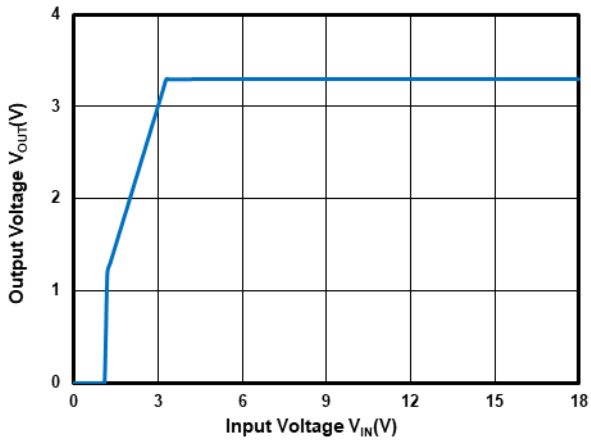
GND Current vs Output Current at V_{OUT}=3.3V



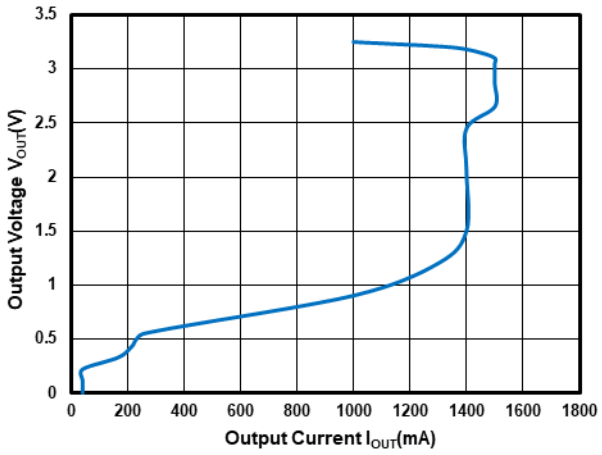
Output Voltage vs Output Current at V_{OUT}=3.3V

■ Typical Performance Characteristics (Continued):

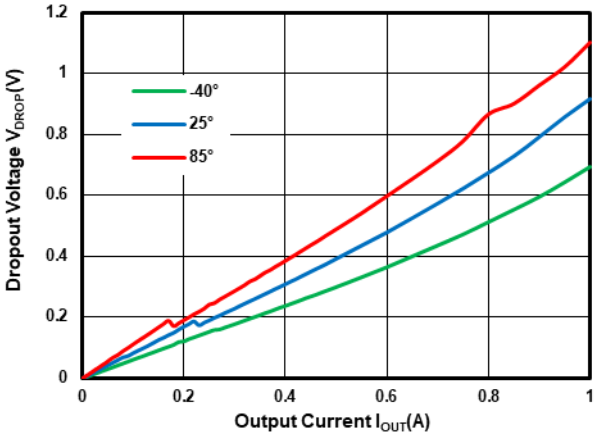
Test Conditions: V_{IN}=V_{OUT}+2.0V, C_{IN}=2.2μF, C_{OUT}=2.2μF, unless otherwise indicated.



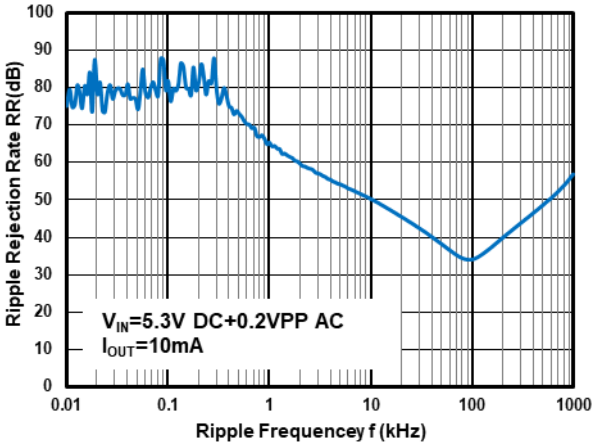
Output Voltage vs Input Voltage at V_{OUT}=3.3V



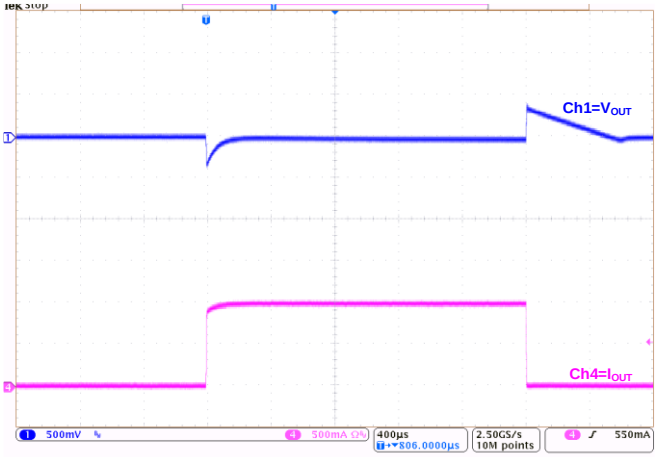
Output Current Fold-back at V_{OUT}=3.3V



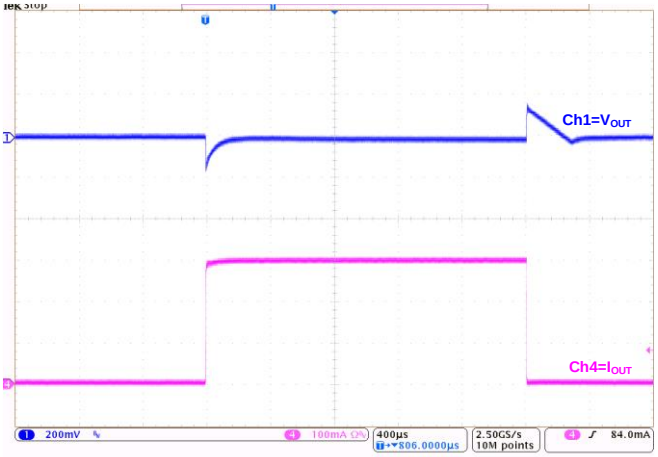
Dropout Voltage vs Temperature at V_{OUT}=3.3V



Power Supply Rejection Ratio at V_{OUT}=3.3V



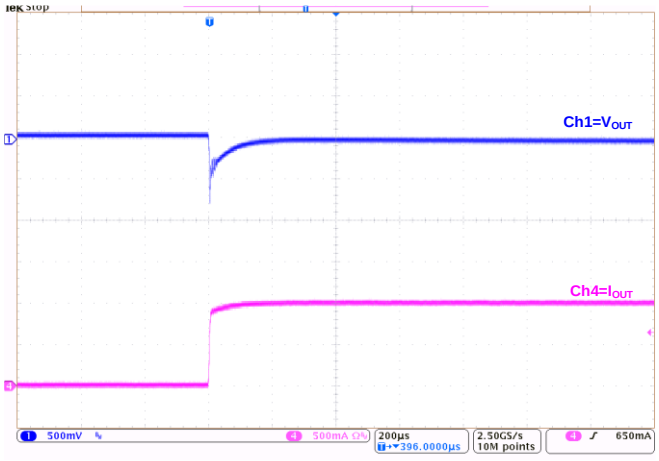
Load Transient at V_{OUT}=3.3V
(I_{OUT}=1mA~1A~1mA)



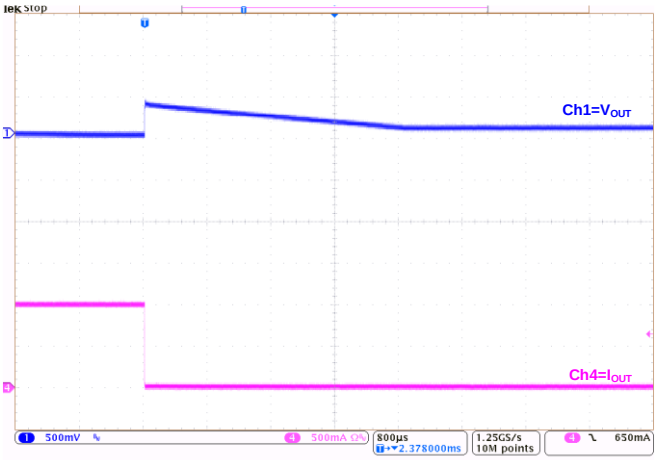
Load Transient at V_{OUT}=3.3V
(I_{OUT}=1mA~300mA~1mA)

■ Typical Performance Characteristics (Continued):

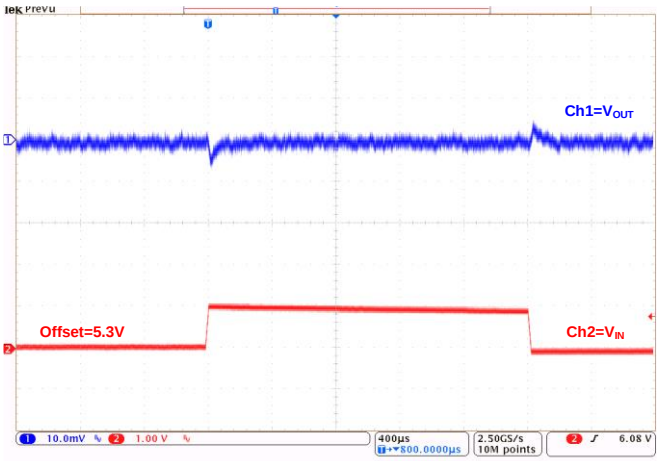
Test Conditions: V_{IN}=V_{OUT}+2.0V, C_{IN}=2.2μF, C_{OUT}=2.2μF, unless otherwise indicated.



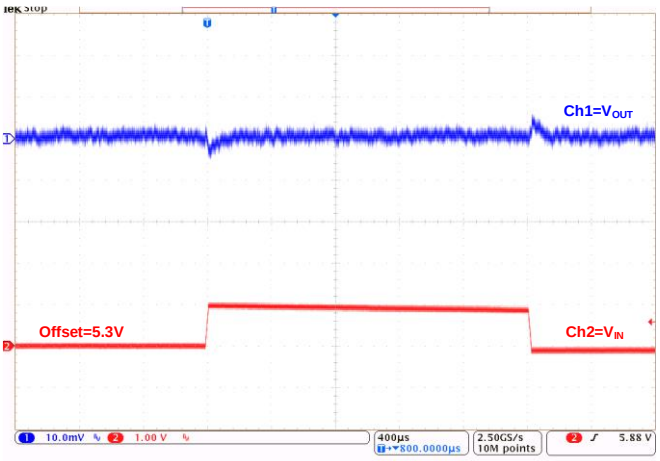
Load Transient at V_{OUT}=3.3V
(I_{OUT}=0mA~1A)



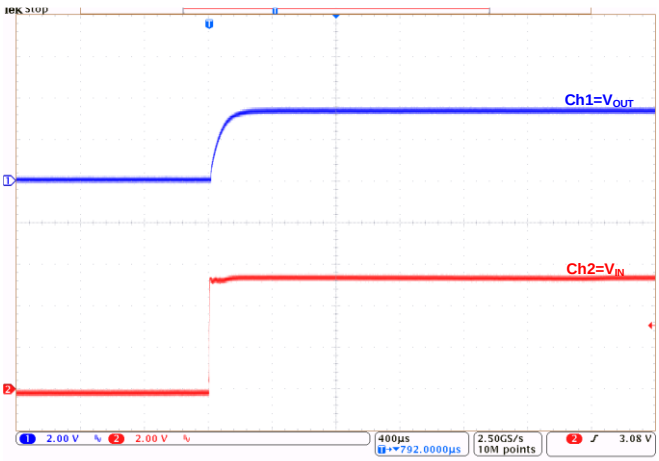
Load Transient at V_{OUT}=3.3V
(I_{OUT}=1A~0mA)



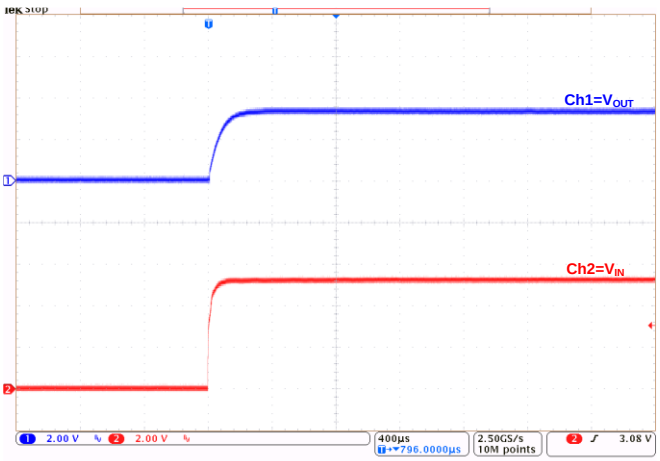
Line Transient at V_{OUT}=3.3V
(I_{OUT}=1mA)



Line Transient at V_{OUT}=3.3V
(I_{OUT}=10mA)



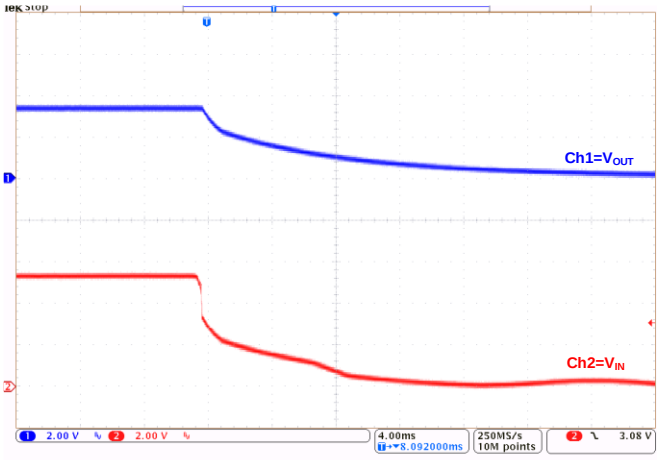
Power-Up at V_{OUT}=3.3V:
(I_{OUT}=1mA)



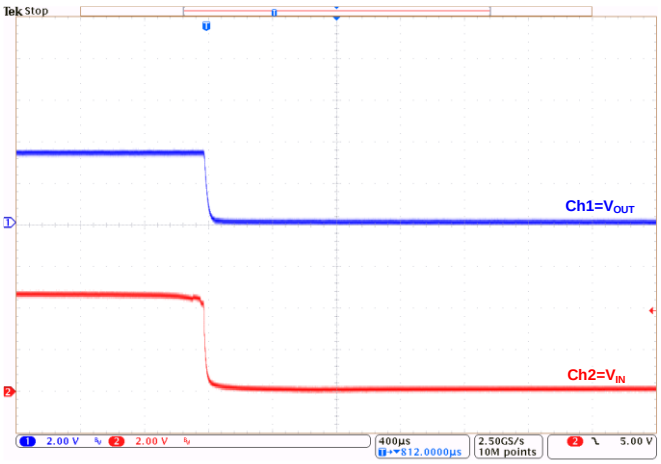
Power-Up at V_{OUT}=3.3V:
(I_{OUT}=1A)

■ Typical Performance Characteristics (Continued):

Test Conditions: V_{IN}=V_{OUT}+2.0V, C_{IN}=2.2μF, C_{OUT}=2.2μF, unless otherwise indicated.



Power-Down at V_{OUT}=3.3V:
(I_{OUT}=1mA))

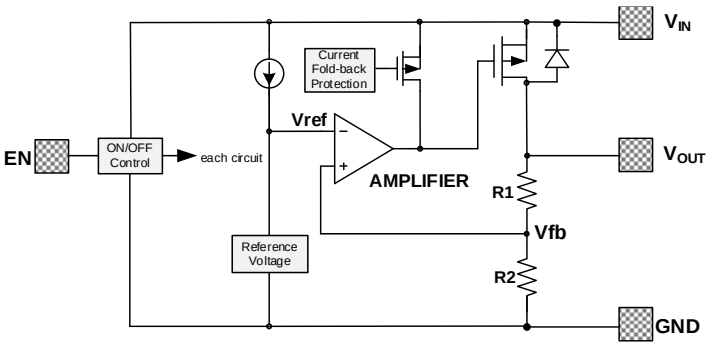


Power-Down at V_{OUT}=3.3V:
(I_{OUT}=1A)

■ Operational Explanation

Output voltage control

The voltage divided by resistors R1 and R2 is compared with the internal reference voltage by the error amplifier. The amplifier output then drives the P-channel MOSFET connected to the V_{OUT} pin. The output voltage at the V_{OUT} pin is regulated by this negative feedback system. The current limit circuit and short protect circuit operate in relation to output current level. Further, the IC's internal circuitry can be in operation or shutdown modes controlled by the CE pin's signal.



■ Pass transistor

The pass transistor with low turn-on resistance used in MD7218 is a P-channel MOSFET. If the potential on V_{OUT} pin is higher than VIN, it is possible that IC will be destroyed due to reverse current which is caused by parasitic diodes between V_{IN} and V_{OUT}. Therefore, the V_{OUT} pin potential exceeds V_{IN}+0.3V is not allowed.

■ Current limit, over temperature protection

The MD7218 series includes a combination of a fixed current limiter circuit which aid the operations of the current limiter and circuit protection. When the load current reaches the current limit level, the fixed current limiter circuit operates and output voltage drops. As a result of this drop in output voltage, the foldback circuit operates, output voltage drops further and output current decreases.

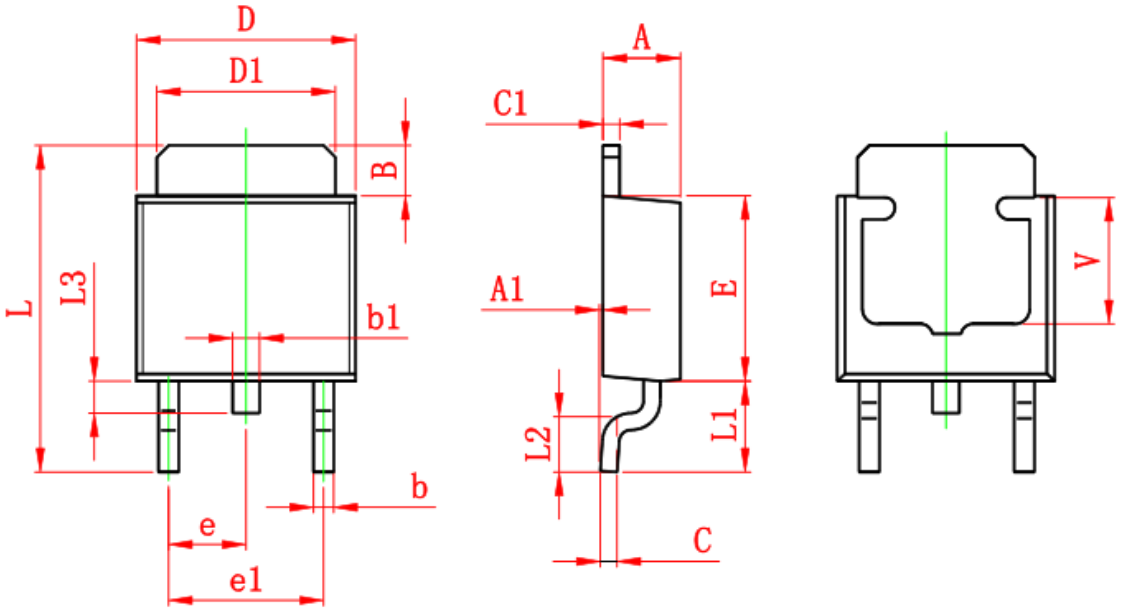
Special attention should be paid to that the product of the dropout voltage on the chip and the output current must be smaller than the heat dissipation. If power consumption on the chip is more than the heat dissipation, OTP will protect the chip from damaging due to over temperature.

■ Notes:

1. The input and output capacitors should be placed as close as possible to the IC.
2. If the impedance of the power supply is high, which is caused by forgetting installing input capacitor or installing too small value capacitor, the oscillation may occur.
3. Pay attention to the operation conditions of input and output voltage and load current, such that the power consumption in the IC should not exceed the allowable power consumption of the package even though the chip has short circuit protection.
4. IC has a built-in anti-static protection (ESD) circuit, but please do not add excessive stress to the IC.

■ Packaging Information

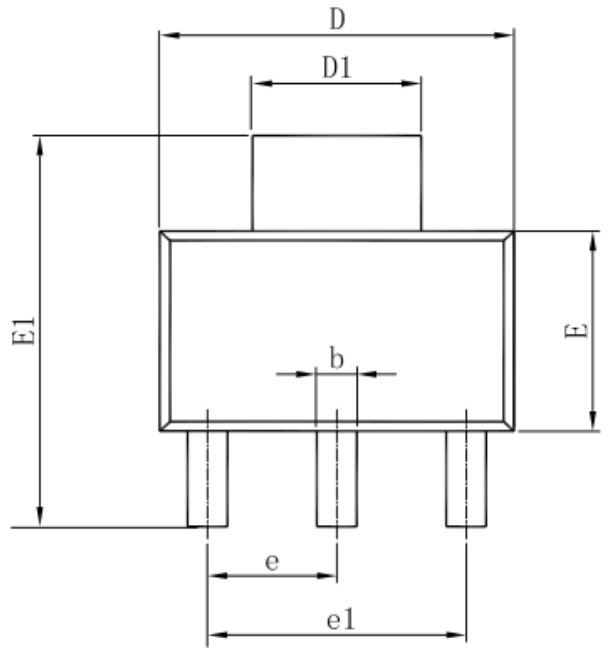
TO-252-2L PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	2.200	2.400	0.087	0.094
A1	0.000	0.127	0.000	0.005
B	1.350	1.650	0.053	0.065
b	0.500	0.700	0.020	0.028
b1	0.700	0.900	0.028	0.035
c	0.430	0.580	0.017	0.023
c1	0.430	0.580	0.017	0.023
D	6.350	6.650	0.250	0.262
D1	5.200	5.400	0.205	0.213
E	5.400	5.700	0.213	0.224
e	2.300 TYP.		0.091 TYP.	
e1	4.500	4.700	0.177	0.185
L	9.500	9.900	0.374	0.390
L1	2.550	2.900	0.100	0.114
L2	1.400	1.780	0.055	0.070
L3	0.600	0.900	0.024	0.035
V	3.800 REF.		0.150 REF.	

■ Packaging Information (Continued)

SOT-223 PACKAGE OUTLINE DIMENSIONS

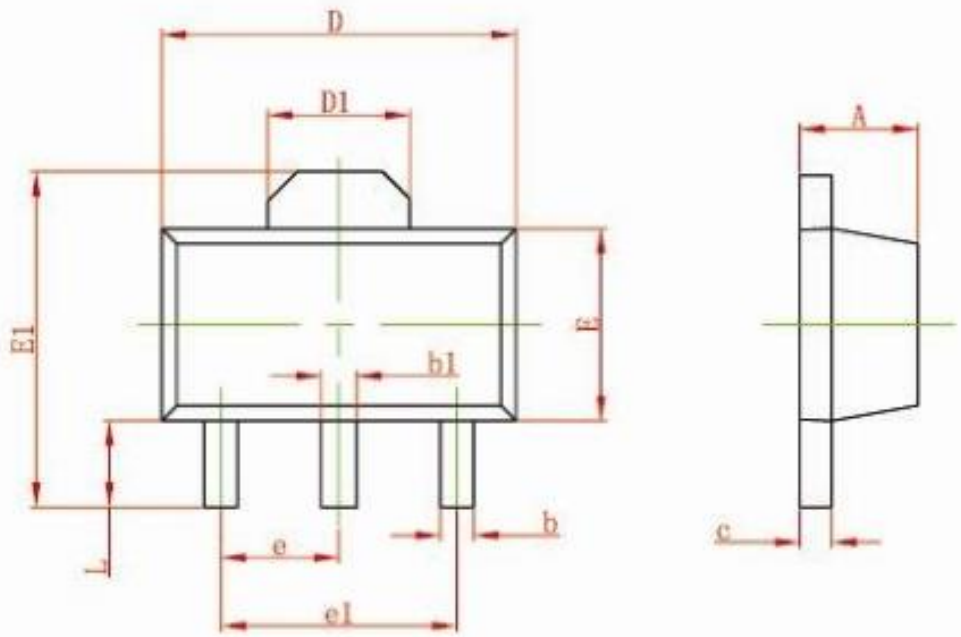




Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.520	1.800	0.060	0.071
A1	0.000	0.100	0.000	0.004
A2	1.500	1.700	0.059	0.067
b	0.660	0.820	0.026	0.032
c	0.250	0.350	0.010	0.014
D	6.200	6.400	0.244	0.252
D1	2.900	3.100	0.114	0.122
E	3.300	3.700	0.130	0.146
E1	6.830	7.070	0.269	0.278
e	2.300(BSC)		0.091(BSC)	
e1	4.500	4.700	0.177	0.185
L	0.900	1.150	0.035	0.045
θ	0°	10°	0°	10°

■ Packaging Information (Continued)

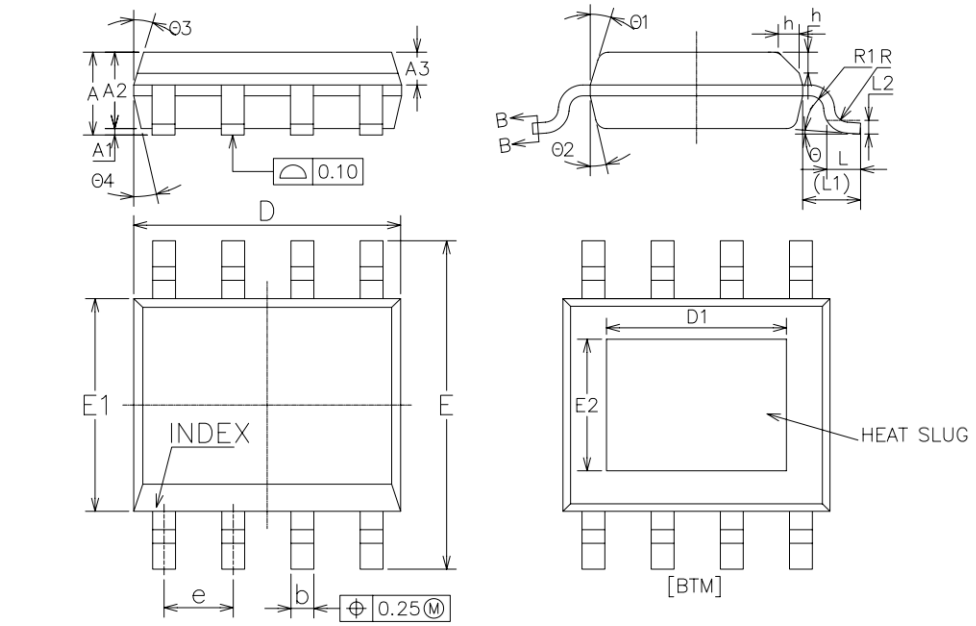
SOT-89-3L PACKAGE OUTLINE DIMENSIONS

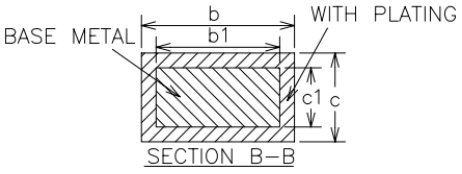


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.197
b1	0.400	0.580	0.016	0.023
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.550 REF		0.061 REF	
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500 TYP		0.060TYP	
e1	3.000 TYP		0.118TYP	
L	0.900	1.200	0.035	0.047

■ Packaging Information (Continued)

ESOP8 PACKAGE OUTLINE DIMENSIONS



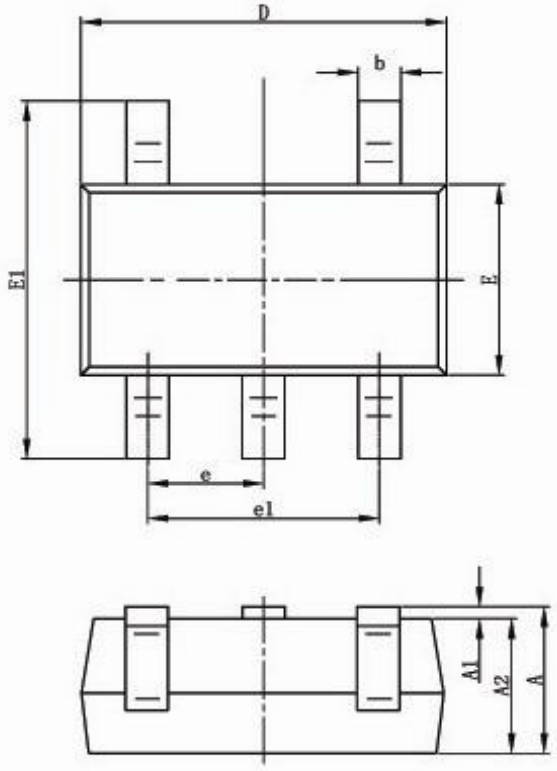


COMMON DIMENSIONS
(UNITS OF MEASURE=MILLIMETER)

SYMBOL	MIN	NOM	MAX
A	1.35	1.55	1.70
A1	0	0.10	0.15
A2	1.25	1.40	1.65
A3	0.50	0.60	0.70
b	0.38	–	0.51
b1	0.37	0.42	0.47
c	0.17	–	0.25
c1	0.17	0.20	0.23
D	4.80	4.90	5.00
D1 Option 1	3.10	3.30	3.50
D1 Option 2	2.09	2.29	2.49
E	5.80	6.00	6.20
E1	3.80	3.90	4.00
E2 Option 1	2.20	2.40	2.60
E2 Option 2	2.09	2.29	2.49
e	1.17	1.27	1.37
L	0.45	0.60	0.80
L1		1.04REF	
L2		0.25BSC	
R	0.07	–	–
R1	0.07	–	–
h	0.30	0.40	0.50
θ	0°	–	8°
θ1	15°	17°	19°
θ2	11°	13°	15°
θ3	15°	17°	19°
θ4	11°	13°	15°

■ Packaging Information (Continued)

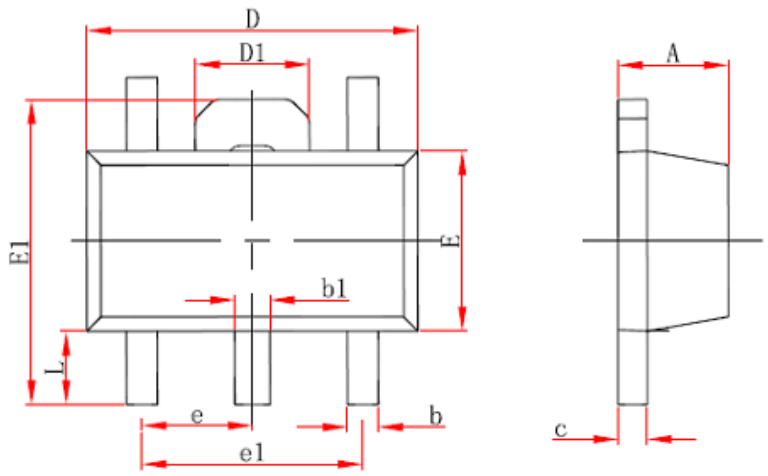
SOT-23-5L PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

■ Packaging Information (Continued)

SOT-89-5L PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.380	0.580	0.015	0.023
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.550 REF.		0.061 REF.	
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500 TYP.		0.060TYP.	
e1	3.000 TYP.		0.118 TYP.	
L	0.900	1.200	0.035	0.047